

HJB Equation & Optimal Stopping

Session 20 • The math behind $L^*(t)$ — when to exit

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1

Why optimal stopping matters

Recap from Session 12

2

The exit decision as a free-boundary problem

Continue or stop

3

HJB equation

Bellman's principle for continuous time

4

Free-boundary conditions

Value matching + smooth pasting

5

Numerical solution preview

How $L^*(t)$ is computed in practice

Why optimal stopping is the right framework

Recall Session 12: the trapped investor's problem.

At every moment, an LP chooses between:

CONTINUE (HOLD)

Keep the position. Receive expected future cash flows minus the time-value of capital.

$$\text{Value} = E[\text{continuation payoff} \mid \text{current } L]$$

STOP (EXIT)

Sell now in the secondary market at the current state-dependent price $\pi(L, T-t)$.

$$\text{Value} = \text{NAV} \cdot (1 - \pi(L, T-t))$$

The exit boundary $L^(t)$ is the value of L at which both options are equally valuable.*

The Hamilton-Jacobi-Bellman equation

Continuation value $V(L,t)$ satisfies the HJB PDE in the continuation region:

$$\frac{\partial V}{\partial t} + \kappa(\bar{L} - L)\frac{\partial V}{\partial L} + \frac{1}{2}\sigma^2\frac{\partial^2 V}{\partial L^2} - r(L)V + CF(L, t) = 0$$

Each term has economic meaning:

$\frac{\partial V}{\partial t}$	Time decay of optionality	$\kappa(\bar{L} - L) \cdot \frac{\partial V}{\partial L}$	Expected change from mean reversion
$\frac{1}{2}\sigma^2 \cdot \frac{\partial^2 V}{\partial L^2}$	Itô correction (uncertainty effect)	$-r(L) \cdot V$	Time-cost of capital
$CF(L,t)$	Cash flows received while holding		

Free-boundary conditions: value matching & smooth pasting

At the boundary $L^*(t)$ where exit becomes optimal, two conditions must hold:

1. VALUE MATCHING

$$V(L^*(t), t) = NAV \cdot (1 - \pi(L^*(t), T - t))$$

Continuation value must equal stop value at the boundary. Otherwise an arbitrage exists.

Economic meaning: 'indifference at the boundary.'

2. SMOOTH PASTING

$$\frac{\partial V}{\partial L} \Big|_{L^*(t)} = \frac{\partial}{\partial L} [NAV \cdot (1 - \pi)] \Big|_{L^*(t)}$$

Derivatives must also match — the value function pastes smoothly at the boundary.

Economic meaning: 'optimal' means no first-order gain from waiting longer.

Together: PDE + boundary conditions determine $L^*(t)$ uniquely.

Session 20 summary

What we accomplished today

- 1 Optimal exit timing = optimal stopping problem with state variable $L(t)$
- 2 Continuation value $V(L,t)$ satisfies the HJB equation in the holding region
- 3 At $L^*(t)$: value matching ($V = \text{stop value}$) + smooth pasting ($\partial V / \partial L = \partial \text{stop} / \partial L$)
- 4 Together: PDE + free-boundary conditions yield $L^*(t)$ uniquely. Numerical methods solve it.

Next session

Session 21: McKean-Vlasov equations and mean-field games — the math behind Failure 3